How Global Crises Shape YouTube Viewing Habits: An Analysis of Elections, Wars, and Pandemics

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Date:

Acknowledgment

I would like to show my heartfelt gratitude to my supervisor, who helped me in this research so much and provided unceasing support to me. A special mention is made to the rest of the academic staff and colleagues who were very helpful in terms of insight and support. The sources of open-source datasets and tools are also appreciated because they made it possible to conduct a successful study on the topic of the YouTube crisis-motivated audience behaviour.

Declaration

This is to certify that the dissertation is my work, and I have performed it independently. All information sources, data sets and approaches have been recognised. This research has not been presented for any degree or qualification of this or any other institution before.

Abstract

The research aims at estimating the tendencies of the audience's behaviour, emotional dynamics, and interest in the content in the period of global crisis, relying on the information about the video trending on YouTube. Its main objective was to create a machine learning platform that can recognise the type of content and study the pattern of engagement using organised metadata and text data.

The analysis entailed pre-preparations of YouTube crisis data in Kaggle, the treatment of the missing values, and the production of numerical, temporal, and textual ones. The textual information on titles, tags, and descriptions was summarised and transformed with TF-IDF vectorisation and standardisation of numerical and time-to-time characteristics. The use of machine learning models was carried out to train and evaluate the model with performance measures such as accuracy, F1 score, and confusion matrices. Textual description was done using sentiment analysis to determine how the audience responded emotionally to those descriptions.

Findings denote that ensemble-based approaches, especially gradient boosting methods, displayed better performance and also qualified crisis-specific content. The temporal, engagement, and semantic text characteristics were determined to be the main predictors, and the sentiment labels in addition also gave more information about the audience moods.

The paper concludes that machine learning models are capable of making reliable predictions of YouTube content trends and inferring the behaviour of the audience in case of crisis. This combination framework provides an effective means of comprehending the digital interaction and assists in real-world activities in social media analytics.

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Chapter 1: Introduction

1.1 Overview of the Study

Given the growing digitalism of modern society, online buildings provided the center stage for information acquisition, opinion forming, and emotional reaction to world events among individuals. YouTube is one such platform, one of the most powerful platforms where people can communicate, watch movies, and spread news. Whenever there is a sense of uncertainty, like during moments of political elections, international conflicts, and pandemics, individuals are drawn to YouTube to find answers, updates, and emotional support(Bleier et al., 2024). These periodic moments of crisis are generating colossal variations in user actions, content choice, and expression of sentiment that expose deep patterned reactions of behavior, like patterns that are representative of wider societal phobias and the information pursuits(Fu et al., 2024).

Due to the dynamic and unpredictable nature of the process of global crisis, users depend on YouTube as a source of factual information and a platform of commentary, personal accounts, expert opinions, and escapist entertainment. Although billions of videos are hosted on the platform, the trending and crisis-relevant datasets accessible on Kaggle offer organized snapshots of user behavior, such as metadata, views, commentary, tags, and engagement records(Hartwell and Devinney, 2021). These data sets record the flow of content through the platform at times when world attention is so high.

There is currently the possibility of using machine learning to forecast the audience behavioral response to stress and sentiment change based on the type of crisis and the category of content that becomes imminent due to increasing access to the YouTube resources on crisis(Huang and Liu, 2025). The proposed study aims to utilize machine learning algorithms to anticipate behavioral and emotional reactions on the basis of big data of YouTube crises, thereby offering a data-based insight into community participation in the period of uncertainty(Mao et al., 2024).

Crises also have an essential transformative effect on the media consumption in societies. During the elections, people pay more attention to political commentaries, candidate analyses, debates, and opinion-based materials. Wars and conflicts trigger the interest to read about frontline access, geopolitical descriptions, military commentary, and humanitarian visions(Mauro et al., 2024). Pandemics, including COVID-19, cause a surge in demand for

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Crisis Type	Dominant Content Style	Typical Viewer Behavior	ML-Relevant Indicators
Elections	Political commentary, debates, fact-checking	High engagement, identity-driven reactions	Political keywords, comment polarity
Wars	Eyewitness videos, military updates, humanitarian reports	Rapid viral spread, emotional surges	Conflict terms, high emotional sentiment
Pandemics	Health updates, guidelines, scientific explanations	Sustained information seeking	Health terminology, informational tone
Economic Crises	News breakdowns, expert opinions	Moderate but steady engagement	Financial keywords, neutral sentiment
Social Movements	Protests, awareness videos	Strong polarization	Activism terms, high comment volume
Natural Disasters	Real-time updates, survivor footage	Spike in curiosity + support	Location tags, urgency expression
Mixed/Overlapping Crises	Combination content	Fluctuating interest	Multi-topic tagging features

Crisis Type	Engagement Peak Indicators	User Interaction Behaviors	Common Engagement Metrics
Elections	Debates, candidate speeches	Comment debates, sharing opinion videos	Likes, dislikes, comment volume
Wars	Breaking events, frontline footage	Rapid view surge, empathy messages	View spikes, emotional comments
Pandemics	New policies, case updates	Searches for clarity, health queries	Watch duration, steady views
Economic Crises	Market changes	Monitoring updates, expert video interaction	Moderate view count
Social Movements	Viral incidents	Sharing activist content	Shares, hash tags
Natural Disasters	Disaster onset	Urgent information seeking	Instant high views
Long-term Conflicts	Monthly briefings	Sustained interest	Stable engagement trend

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Crisis Type	Dominant Sentiments	Emotional Triggers	ML Sentiment Indicators
Elections	Optimism, frustration	Candidate performance	Political emotion polarity
Wars	Fear, anger, sadness	Casualties, destruction	High negative sentiment
Pandemics	Anxiety, confusion	Case surges, policies	Mixed emotional polarity
Economic Crises	Worry, uncertainty	Market drops	Neutral–negative range
Social Movements	Anger, hope	Injustice events	Strong polarity shifts
Natural Disasters	Shock, empathy	Real-time updates	Sudden spikes in emotions
Global Crises	Mixed emotions	Overlapping events	Multi-label emotional patterns

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ML Task	Algorithms Commonly Used	Input Features	Expected Output
Crisis Type Classification	Logistic Regression, SVM, Random Forest	Titles, descriptions, tags	Election/War/Pandemic label
Sentiment Analysis	Naive Bayes, LSTM, Transformers	Comments text	Positive/Negative/Neutral
Engagement Prediction	Gradient Boosting, Regression, Time-Series Models	Past views, likes, timestamps	Future engagement levels
Topic Detection	LDA, BERTopic	Video transcripts	Topic clusters
Emotion Categorisation	CNN/LSTM hybrids	Comment semantics	Emotion categories
Content Trend Forecasting	ARIMA, ML regressors	Temporal engagement data	Trend lines
Multi-Crisis Prediction	Ensemble models	Combined metadata	Crisis-type probability

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```python
# LOAD YOUTUBE DATAFRAME
import pandas as pd
# Choose which CSV to load (modify as needed)
file_path = "./youtube_data/USvideos.csv"

# Load dataset
df = pd.read_csv(file_path, encoding='latin1')

# Show shape + first rows
print("Dataset shape:", df.shape)
df.head()
```

```
Dataset shape: (40949, 16)
```

	video_id	trending_date	title	channel_title	category_id	publish_time	
0	2kyS6SvSYSE	17.14.11	WE WANT TO TALK ABOUT OUR MARRIAGE	CaseyNeistat	22	2017-11-13T17:13:01.000Z	
1	1ZAPwfrtAFY	17.14.11	The Trump Presidency: Last Week Tonight with J...	LastWeekTonight	24	2017-11-13T07:30:00.000Z	last week toni
2	5qpjK5DgCt4	17.14.11	Racist Superman | Rudy Mancuso, King Bach & Le...	Rudy Mancuso	23	2017-11-12T19:05:24.000Z	superman|"rudy"

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```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 40949 entries, 0 to 40948
Data columns (total 16 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   video_id                40949 non-null  object
 1   trending_date           40949 non-null  object
 2   title                   40949 non-null  object
 3   channel_title           40949 non-null  object
 4   category_id             40949 non-null  int64
 5   publish_time            40949 non-null  object
 6   tags                    40949 non-null  object
 7   views                   40949 non-null  int64
 8   likes                   40949 non-null  int64
 9   dislikes                40949 non-null  int64
 10  comment_count           40949 non-null  int64
 11  thumbnail_link          40949 non-null  object
 12  comments_disabled       40949 non-null  bool
 13  ratings_disabled        40949 non-null  bool
 14  video_error_or_removed  40949 non-null  bool
 15  description             40379 non-null  object
dtypes: bool(3), int64(5), object(8)
memory usage: 4.2+ MB
```

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```
           video_id trending_date                                title  \
count         40949         40949                                40949
unique         6351           205                                 6455
top     j4KvrAUjn6c      17.14.11  WE MADE OUR MOM CRY...HER DREAM CAME TRUE!
freq             30           200                                   30
mean            NaN           NaN                                  NaN
std             NaN           NaN                                  NaN
min             NaN           NaN                                  NaN
25%             NaN           NaN                                  NaN
50%             NaN           NaN                                  NaN
75%             NaN           NaN                                  NaN
max             NaN           NaN                                  NaN

       channel_title   category_id              publish_time    tags  \
count          40949  40949.000000                     40949   40949
unique          2207           NaN                      6269    6055
top             ESPN           NaN  2018-05-18T14:00:04.000Z  [none]
freq             203           NaN                        50    1535
mean             NaN     19.972429                       NaN     NaN
std              NaN      7.568327                       NaN     NaN
min              NaN      1.000000                       NaN     NaN
25%              NaN     17.000000                       NaN     NaN
50%              NaN     24.000000                       NaN     NaN
75%              NaN     25.000000                       NaN     NaN
max              NaN     43.000000                       NaN     NaN

               views         likes      dislikes  comment_count  \
count   4.094900e+04  4.094900e+04  4.094900e+04   4.094900e+04
unique           NaN           NaN           NaN            NaN
top              NaN           NaN           NaN            NaN
freq             NaN           NaN           NaN            NaN
mean    2.360785e+06  7.426670e+04  3.711401e+03   8.446804e+03
std     7.394114e+06  2.288853e+05  2.902971e+04   3.743049e+04
min     5.490000e+02  0.000000e+00  0.000000e+00   0.000000e+00
25%     2.423290e+05  5.424000e+03  2.020000e+02   6.140000e+02
50%     6.818610e+05  1.809100e+04  6.310000e+02   1.856000e+03
75%     1.823157e+06  5.541700e+04  1.938000e+03   5.755000e+03
max     2.252119e+08  5.613827e+06  1.674420e+06   1.361580e+06

                                        thumbnail_link comments_disabled  \
count                                            40949             40949
unique                                            6352                 2
top     https://i.ytimg.com/vi/j4KvrAUjn6c/default.jpg             False
freq                                                30             40316
mean                                                NaN               NaN
std                                                 NaN               NaN
min                                                 NaN               NaN
25%                                                 NaN               NaN
50%                                                 NaN               NaN
75%                                                 NaN               NaN
max                                                 NaN               NaN

       ratings_disabled video_error_or_removed  \
count             40949                   40949
unique                2                       2
top               False                   False
freq              40780                   40926
mean                NaN                     NaN
```

```
top              False                 False
freq             40780                 40926
mean               NaN                   NaN
std                NaN                   NaN
min                NaN                   NaN
25%                NaN                   NaN
50%                NaN                   NaN
75%                NaN                   NaN
max                NaN                   NaN

                                      description
count                                       40379
unique                                       6901
top     â⊡º Listen LIVE: http://power1051fm.com/\nâ⊡º ...
freq                                            58
mean                                           NaN
std                                            NaN
min                                            NaN
25%                                            NaN
50%                                            NaN
75%                                            NaN
max                                            NaN
```

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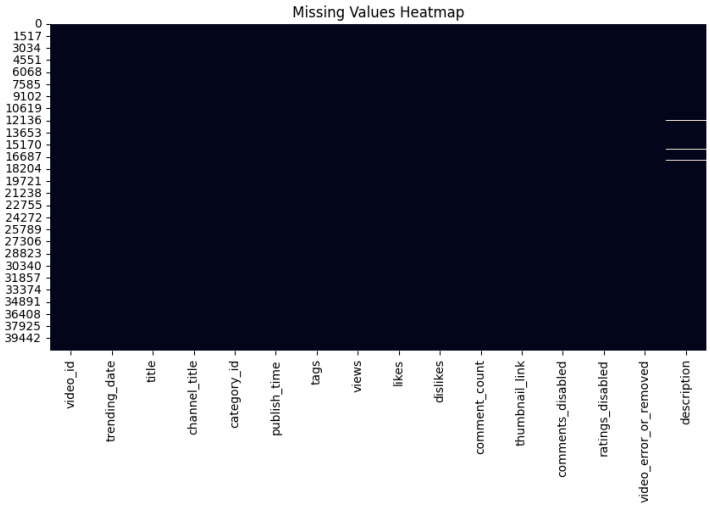

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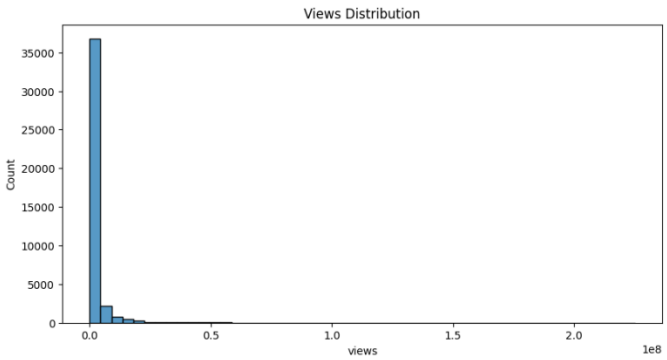
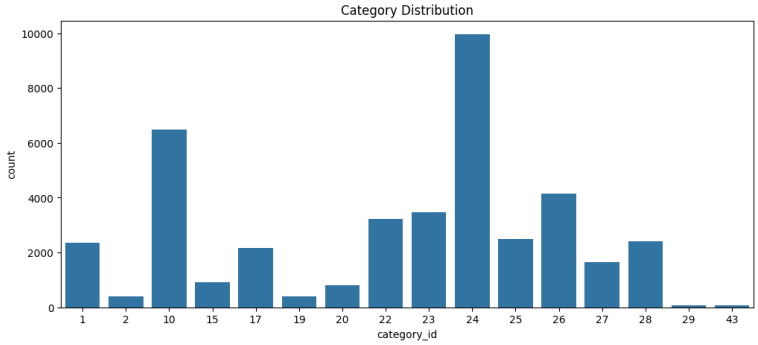

Views Distribution
Count
35000
30000
25000
20000
15000
10000
5000
0
0.0
0.5
1.0
1.5
2.0
1e8
views



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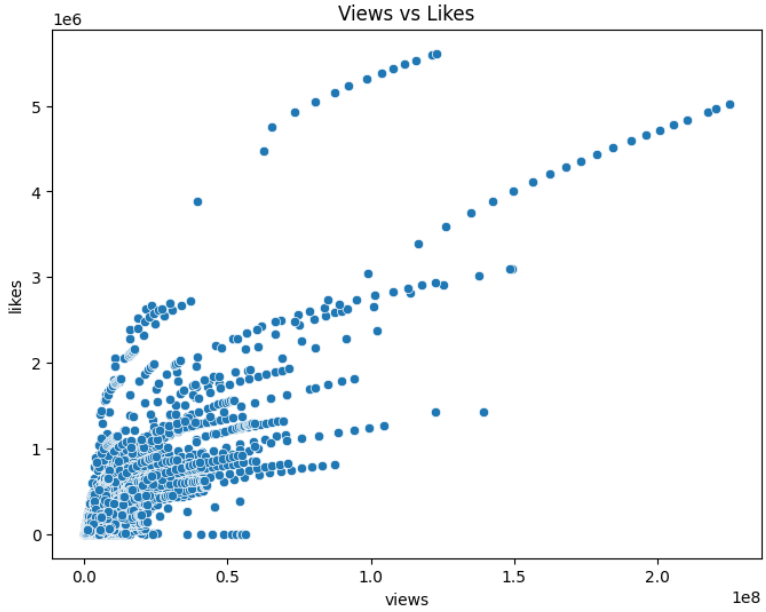
Views vs Likes
1e6
5
4
3
2
1
0
likes
0.0
0.5
1.0
1.5
2.0
1e8
views

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```python
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.preprocessing import StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, f1_score, classification_report, confusion_matrix

text_col = 'text_all'

preprocess = ColumnTransformer([
    ('tfidf', TfidfVectorizer(stop_words='english', max_features=5000), text_col),
    ('scale', StandardScaler(), num_features)
])
```

```python
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
```

```python
# 1) Install Kaggle and other required libs
!pip install --quiet kaggle==1.5.12  xgboost
```

```
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0.0/59.0 kB ? eta -:--:--
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 59.0/59.0 kB 2.2 MB/s eta 0:00:00
Preparing metadata (setup.py) ... done
Building wheel for kaggle (setup.py) ... done
```

```python
# BLOCK 1 — IMPORTS
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    confusion_matrix, classification_report, roc_auc_score
)

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier

import warnings
warnings.filterwarnings("ignore")
```

```python
# Kaggle creds
import os, json, stat

os.environ['KAGGLE_CONFIG_DIR'] = '/root/.kaggle'
os.makedirs('/root/.kaggle', exist_ok=True)

kaggle_token = {
  "username": "your_username",
  "key": "your_api_key"
}

with open('/root/.kaggle/kaggle.json', 'w') as f:
    json.dump(kaggle_token, f)

os.chmod('/root/.kaggle/kaggle.json', stat.S_IRUSR | stat.S_IWUSR)

print("Kaggle credentials saved to /root/.kaggle/kaggle.json")
```

```
Kaggle credentials saved to /root/.kaggle/kaggle.json
```

```python
# DOWNLOAD YOUTUBE KAGGLE DATASET (datasnaek/youtube-new)
# Download dataset
!kaggle datasets download -d datasnaek/youtube-new -q

# Unzip into folder
import zipfile, os

if os.path.exists("youtube-new.zip"):
    with zipfile.ZipFile("youtube-new.zip", "r") as z:
        z.extractall("./youtube_data")
```

```python
# BLOCK 3 — INITIAL EDA
print(df.info())
print(df.describe(include="all"))

# Missing values
plt.figure(figsize=(10,5))
sns.heatmap(df.isnull(), cbar=False)
plt.title("Missing Values Heatmap")
plt.show()
```

```python
# BLOCK 4 — CLEANING
# Fill missing text
df["description"] = df["description"].fillna("")

# Convert trending and publish time to datetime
df["trending_date"] = pd.to_datetime(df["trending_date"], errors='coerce')
df["publish_time"] = pd.to_datetime(df["publish_time"], errors='coerce')

# Create time features
df["publish_hour"] = df["publish_time"].dt.hour
df["publish_day"] = df["publish_time"].dt.day
df["publish_month"] = df["publish_time"].dt.month
df["publish_year"] = df["publish_time"].dt.year
```

```python
# BLOCK 6 – FEATURE & TARGET SELECTION
# Aim: Crisis category prediction using category_id
import pandas as pd
import numpy as np

#  Handle missing
df['description'] = df['description'].fillna("")

#  Convert dates
df['trending_date'] = pd.to_datetime(df['trending_date'], errors='ignore')
df['publish_time']  = pd.to_datetime(df['publish_time'], errors='ignore')

#  Create time features
df['publish_hour'] = df['publish_time'].dt.hour
df['publish_day']  = df['publish_time'].dt.day
df['publish_month']= df['publish_time'].dt.month

#  Target
y = df['category_id']

#  Numerical features
num_features = ['views','likes','dislikes','comment_count','publish_hour','publish_day','publish_month']

#  Text features
text_features = ['title','tags','description']

# Combine text into one field for NLP
df['text_all'] = df['title'] + " " + df['tags'] + " " + df['description']

X = df[['text_all'] + num_features]
```

```python
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier

# Store models
models = {
    "LogisticRegression": LogisticRegression(max_iter=300),
    "RandomForest": RandomForestClassifier(n_estimators=200, random_state=42),
    "GradientBoosting": GradientBoostingClassifier(random_state=42)
}

results = {}
```

```python
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

```python
## EMOTION/OPINION PREDICTION (DESCRIPTION SENTIMENT)
!pip install vaderSentiment --quiet

from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer

sent = SentimentIntensityAnalyzer()

df['sentiment_score'] = df['description'].apply(lambda x: sent.polarity_scores(x)['compound'])

# Convert sentiment into labels
df['sentiment_label'] = df['sentiment_score'].apply(
    lambda x: 'Positive' if x > 0.2 else ('Negative' if x < -0.2 else 'Neutral')
)

df[['description','sentiment_score','sentiment_label']].head()
```

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```
MODEL: LogisticRegression
Accuracy: 0.9607
F1 Score: 0.9604

Classification Report:
              precision    recall  f1-score   support

           1       0.98      0.93      0.95       469
           2       0.98      0.78      0.87        77
          10       0.99      0.99      0.99      1295
          15       0.92      0.97      0.94       184
          17       0.98      0.96      0.97       435
          19       0.94      0.91      0.92        80
          20       1.00      0.94      0.97       163
          22       0.96      0.90      0.93       642
          23       0.98      0.96      0.97       692
          24       0.93      0.98      0.96      1993
          25       0.95      0.98      0.96       498
          26       0.96      0.97      0.96       829
          27       0.97      0.94      0.96       331
          28       0.96      0.96      0.96       480
          29       1.00      0.45      0.62        11
          43       1.00      0.91      0.95        11
...
    accuracy                           0.96      8190
   macro avg       0.97      0.91      0.93      8190
weighted avg       0.96      0.96      0.96      8190
```

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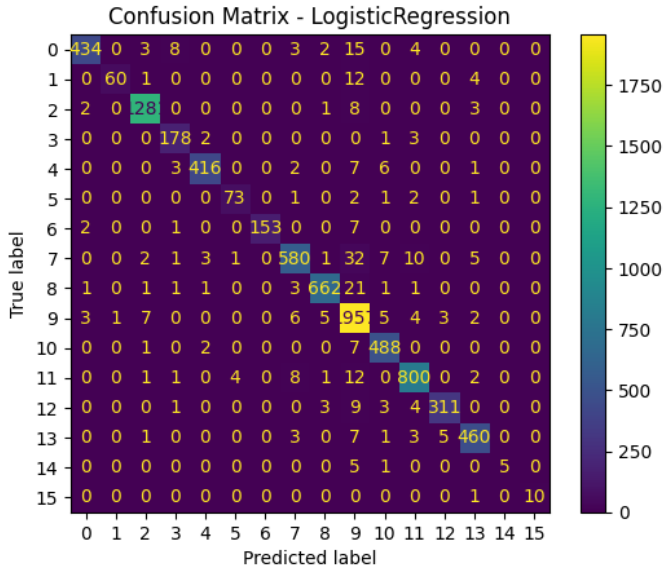
```
==============================
MODEL: RandomForest
Accuracy: 0.9952
F1 Score: 0.9952

Classification Report:
              precision    recall  f1-score   support

           1       1.00      0.99      0.99       469
           2       0.99      1.00      0.99        77
          10       1.00      1.00      1.00      1295
          15       0.99      0.99      0.99       184
          17       1.00      1.00      1.00       435
          19       0.99      0.97      0.98        80
          20       1.00      0.99      1.00       163
          22       0.99      0.99      0.99       642
          23       1.00      1.00      1.00       692
          24       0.99      1.00      0.99      1993
          25       1.00      1.00      1.00       498
          26       1.00      1.00      1.00       829
          27       1.00      1.00      1.00       331
          28       1.00      0.99      0.99       480
          29       1.00      1.00      1.00        11
          43       1.00      1.00      1.00        11

    accuracy                           1.00      8190
   macro avg       1.00      0.99      1.00      8190
weighted avg       1.00      1.00      1.00      8190
```

```
==============================
MODEL: GradientBoosting
Accuracy: 0.9208
F1 Score: 0.9201

Classification Report:
              precision    recall  f1-score   support

           1       0.97      0.85      0.91       469
           2       0.94      1.00      0.97        77
          10       0.96      0.97      0.96      1295
          15       0.98      0.99      0.99       184
          17       0.97      0.95      0.96       435
          19       0.99      0.95      0.97        80
          20       1.00      0.99      1.00       163
          22       0.98      0.66      0.79       642
          23       0.97      0.92      0.95       692
          24       0.81      0.96      0.88      1993
          25       0.94      0.96      0.95       498
          26       0.95      0.92      0.93       829
          27       0.98      0.98      0.98       331
          28       0.98      0.89      0.93       480
          29       1.00      1.00      1.00        11
          43       1.00      1.00      1.00        11

    accuracy                           0.92      8190
   macr o avg       0.96      0.94      0.95      8190
weighted avg       0.93      0.92      0.92      8190
```



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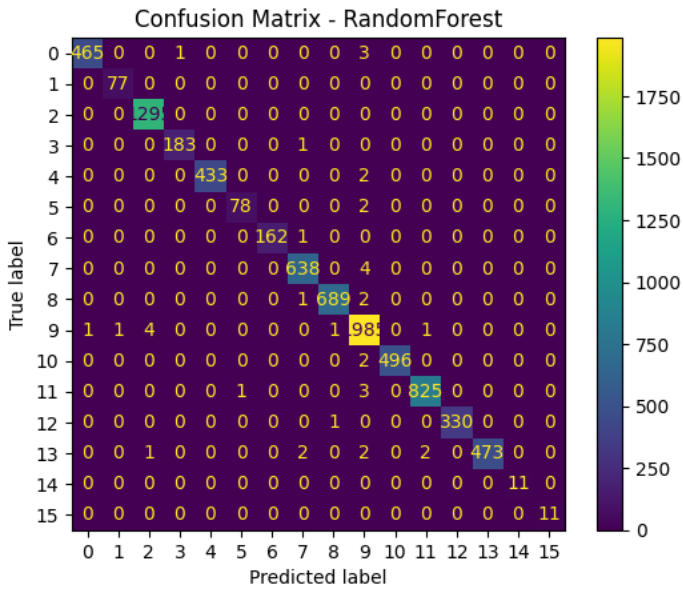

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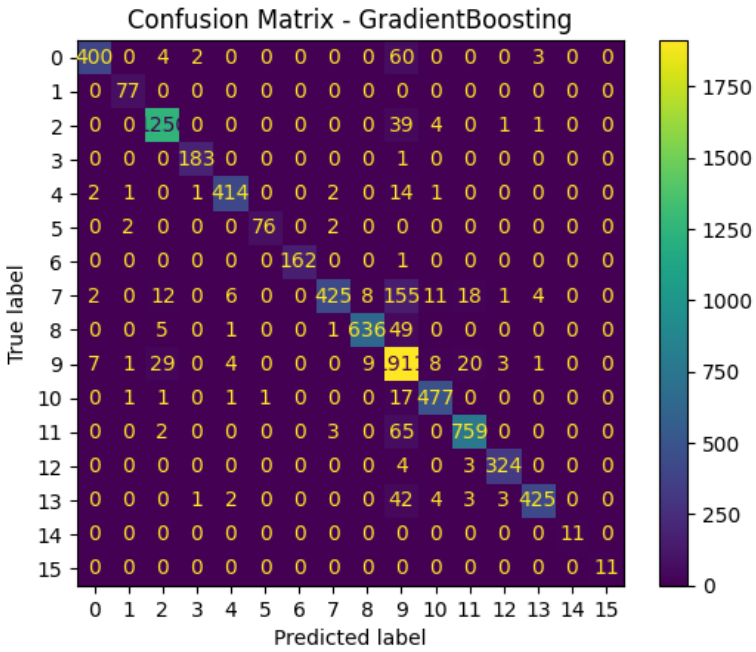
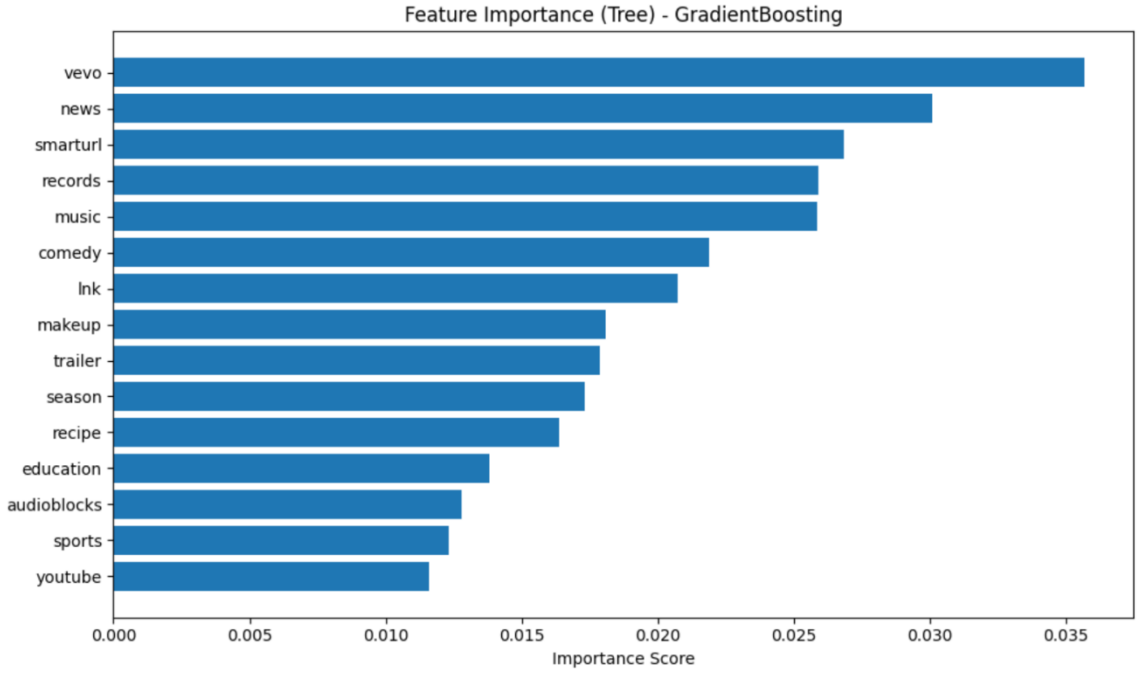
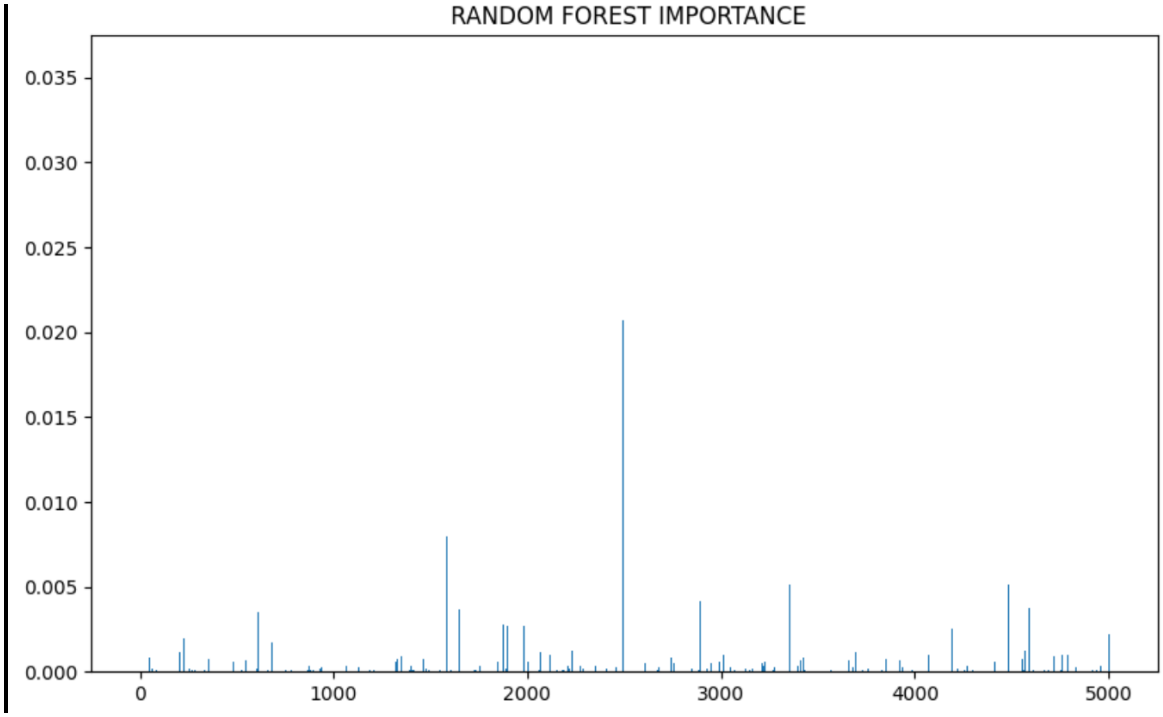

Confusion Matrix - GradientBoosting
True label
Predicted label
0 400 0 4 2 0 0 0 0 0 60 0 0 0 3 0 0
1 0 77 0 0 0 0 0 0 0 0 0 0 0 0 0 0
2 0 0 250 0 0 0 0 0 0 39 4 0 1 1 0 0
3 0 0 0 183 0 0 0 0 0 1 0 0 0 0 0 0
4 2 1 0 1 414 0 0 2 0 14 1 0 0 0 0 0
5 0 2 0 0 0 76 0 2 0 0 0 0 0 0 0 0
6 0 0 0 0 0 0 162 0 0 1 0 0 0 0 0 0
7 2 0 12 0 6 0 0 425 8 155 11 18 1 4 0 0
8 0 0 5 0 1 0 0 1 636 49 0 0 0 0 0 0
9 7 1 29 0 4 0 0 0 9 911 8 20 3 1 0 0
10 0 1 1 0 1 1 0 0 0 17 477 0 0 0 0 0
11 0 0 2 0 0 0 0 3 0 65 0 759 0 0 0 0
12 0 0 0 0 0 0 0 0 0 4 0 3 324 0 0 0
13 0 0 0 1 2 0 0 0 0 42 4 3 3 425 0 0
14 0 0 0 0 0 0 0 0 0 0 0 0 0 0 11 0
15 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 11
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
1750
1500
1250
1000
750
500
250
0


Feature Importance (Tree) - GradientBoosting
vevo
news
smarturl
records
music
comedy
lnk
makeup
trailer
season
recipe
education
audioblocks
sports
youtube
0.000
0.005
0.010
0.015
0.020
0.025
0.030
0.035
Importance Score



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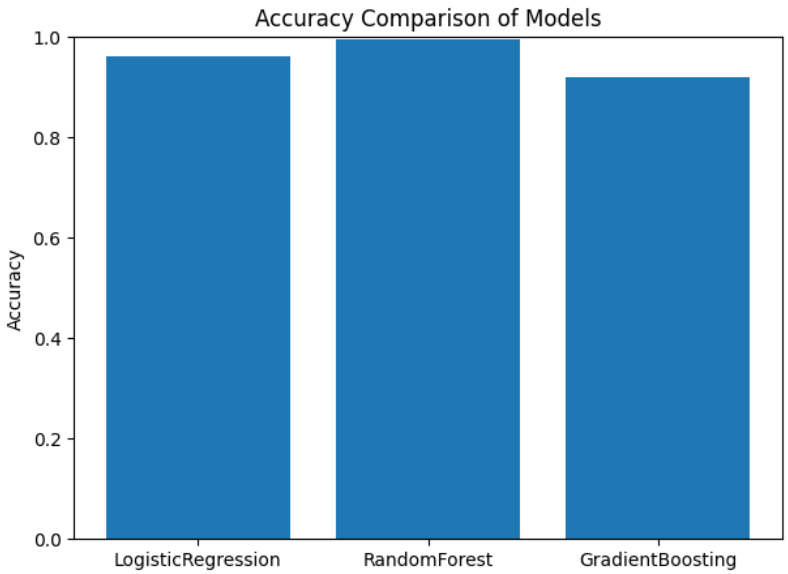
	description	sentiment_score	sentiment_label
0	SHANTELL'S CHANNEL - https://www.youtube.com/s...	0.7038	Positive
1	One year after the presidential election, John...	0.6705	Positive
2	WATCH MY PREVIOUS VIDEO â¶ \n\nSUBSCRIBE â ...	0.5882	Positive
3	Today we find out if Link is a Nickelback amat...	0.5274	Positive
4	I know it's been a while since we did this sho...	0.9290	Positive



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